



DIGITAL LOGIC

Chapters 6 and 7

Analysis and Design of Asynchronous Logic Circuits

MSI General-purpose Circuits

Dr. Fei Chao

School of Information Science and Engineering

Xiamen University

fchao@xmu.edu.cn

Pulse asynchronous sequential logic

There is no uniform clock signal in the asynchronous sequential logic circuit, and the change of the circuit state is the direct result of the external input signal change.

According to the circuit structure and the form of the input signal, the asynchronous sequential logic circuit can be divided into two types: pulse asynchronous sequential logic circuit and level asynchronous sequential logic circuit.

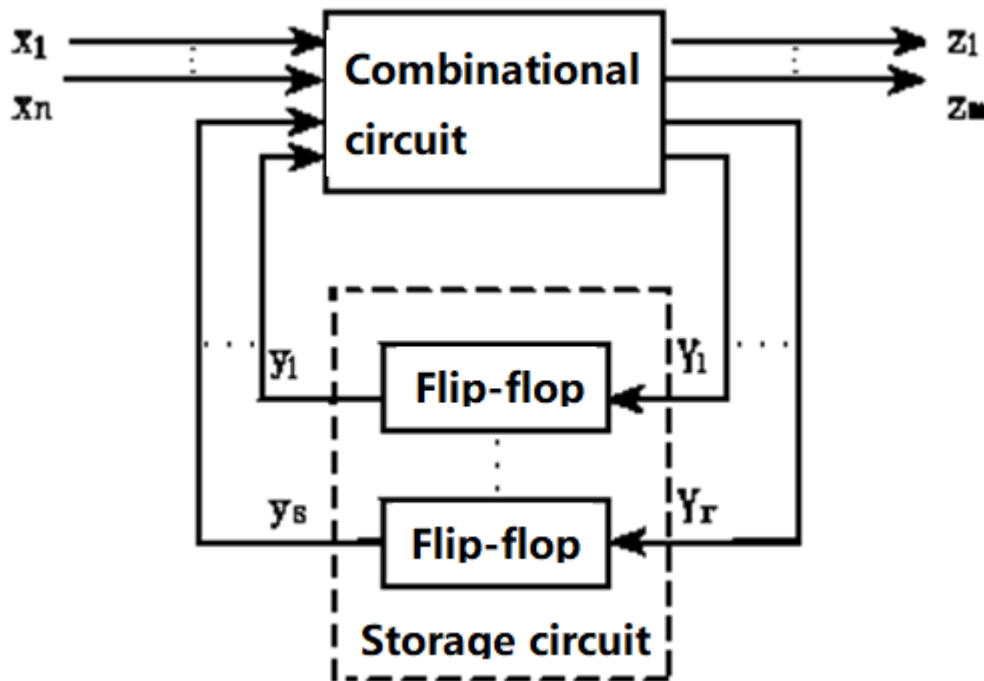
Both types of circuits have two structural models, Mealy and Moore.

6.1 Pulse Asynchronous Sequential Logic Circuit

6.1.1 Introduction

I. Structure

The general structure of the pulse asynchronous sequential circuit is shown in the figure below.



In the figure, the memory circuit can be composed of a clocked control flip-flop or a non-clocked control flip-flop.

II. the form and constraints of the input signal

1. The input signal is the pulse signal;
2. The width of the input pulse must ensure that the flip-flop is flipped reliably;
3. The duration between two adjacent input pulses must ensure that the circuit's reaction can be completed before the pulse arrives;
4. It is not allowed to have pulse signals at the same time for two or more inputs.

For a circuit with n inputs, each time, only $n+1$ input combinations are allowed, and the effective input combination is n .

III. Form of the output signal



The output signal of the pulse asynchronous sequential logic circuit may be a pulse signal or a level signal.

6.1.2 Analysis of Pulse Asynchronous Sequential Logic Circuit

I. Analysis methods and steps

1. Analysis methods

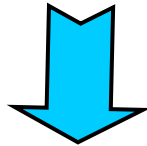
The analysis method of the pulse asynchronous sequential logic circuit is substantially the same as the synchronous sequential logic circuit.

Note that:

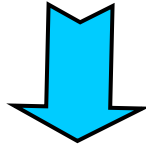
- (1) When the storage element uses a clock control flip-flop, the clock control input of the flip-flop should be treated as an excitation function. The state transition direction is determined according to the input of the flip-flop only when there is a pulse on the clock input. Otherwise, the flip-flop state does not change.**
- (2) According to the constraints on the input, the analysis can eliminate the occurrence of simultaneous pulses on two or more input terminals and no pulse on the input, thus simplifying the diagram and table.**

2. Analysis steps

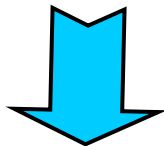
(1) Write the output function and excitation function expression of the circuit;



(2) List circuit next state truth table or excitation equations;



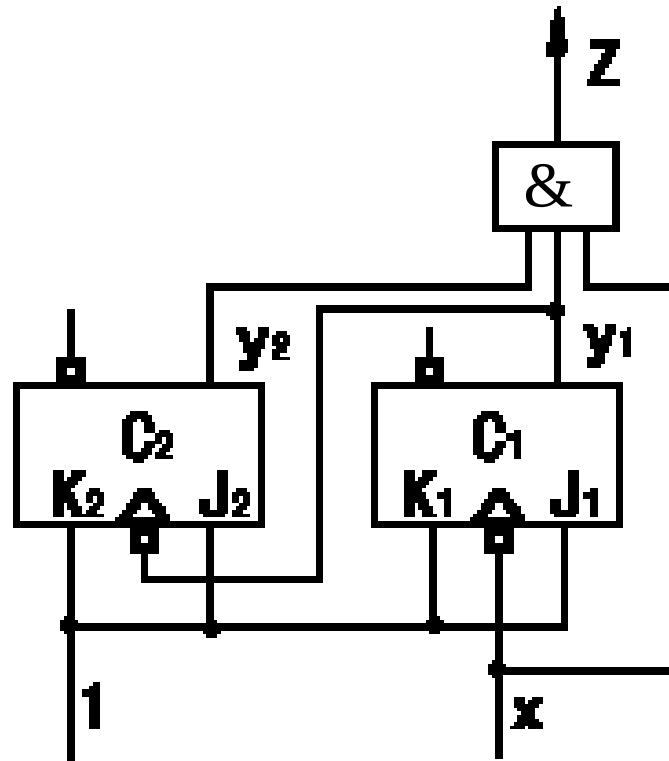
(3) Make a state table and a state diagram;

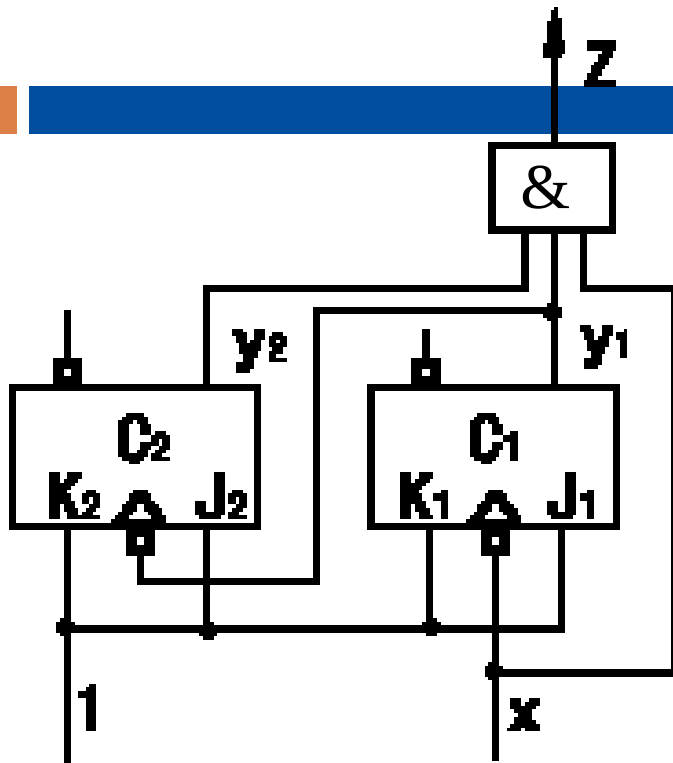


(4) Describe the logic function of the circuit in words (draw a timing diagram if necessary)。

II. Example

Example: Analyze the pulse asynchronous sequential logic circuit shown in the figure below and indicate the function of the circuit.





Solution: The circuit consists of two J-K flip-flops and an AND gate. There is an input terminal x and an output terminal Z . The output is a function of the input and state, and belongs to the Mealy type pulse asynchronous sequential circuit.

Write out the output function and the excitation function expression

$$Z = xy_2y_1$$

$$J_2 = K_2 = 1 \quad ; \quad C_2 = y_1$$

$$J_1 = K_1 = 1 \quad ; \quad C_1 = x$$

(2) List circuit next state truth table

The state transition of the J-K flip-flop occurs at the moment when the clock terminal pulse is in the negative transition. To emphasize the flip-flop changes in the negative transition of C_1 and C_2 , so, use the “ $\downarrow$ ” to present the negative transition. Only when the CP has “ $\downarrow$ ”, the flip-flop can change its state.

Input	Present	Excitation						
x	$y_2 y_1$	J_2	K_2	C_2	J_1	K_1	C_1	$y_2^{(n+1)} y_1^{(n+1)}$
1	0 0	1	1	$\downarrow$	1	1	$\downarrow$	0 1
1	0 1	1	1	$\downarrow$	1	1	$\downarrow$	1 0
1	1 0	1	1	$\downarrow$	1	1	$\downarrow$	1 1
1	1 1	1	1	$\downarrow$	1	1	$\downarrow$	0 0

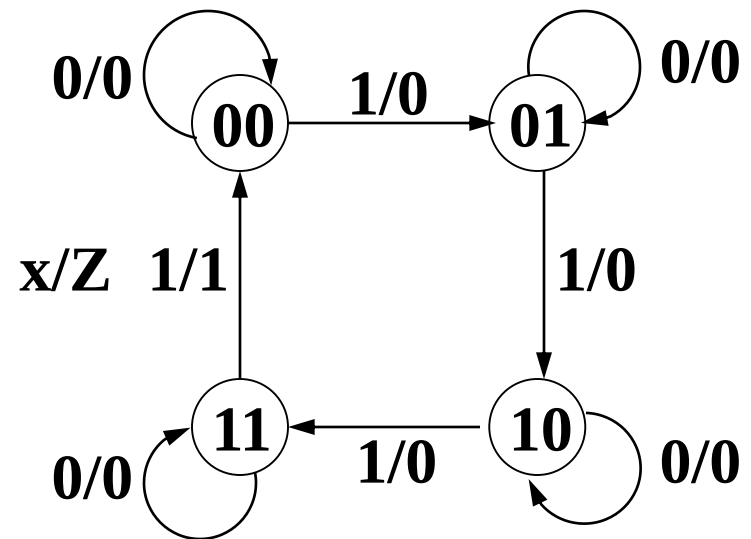
J	K	$Q^{(n+1)}$
0	0	Q
0	1	0
1	0	1
1	1	$\overline{Q}$



(3) List the state table and state diagram

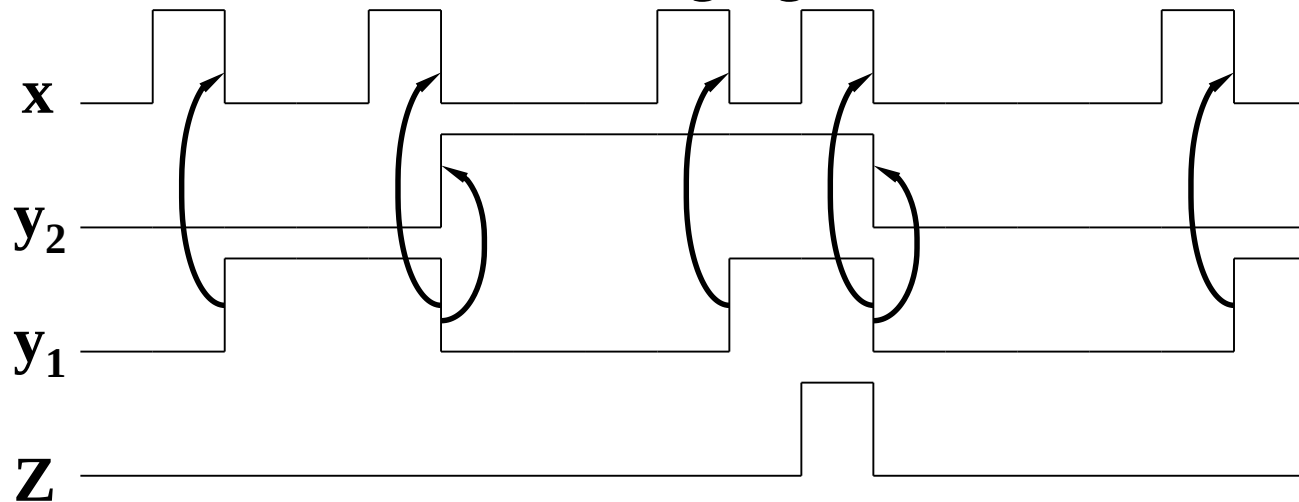
According to the next state truth table and the output function expression ($Z = xy_2y_1$), the circuit's state table and state diagram are shown as:

Present	Next $y_2^{n+1}y_1^{n+1}$ / Output Z
$y_2 y_1$	$x = 1$
0 0	0 1 / 0
0 1	1 0 / 0
1 0	1 1 / 0
1 1	0 0 / 1



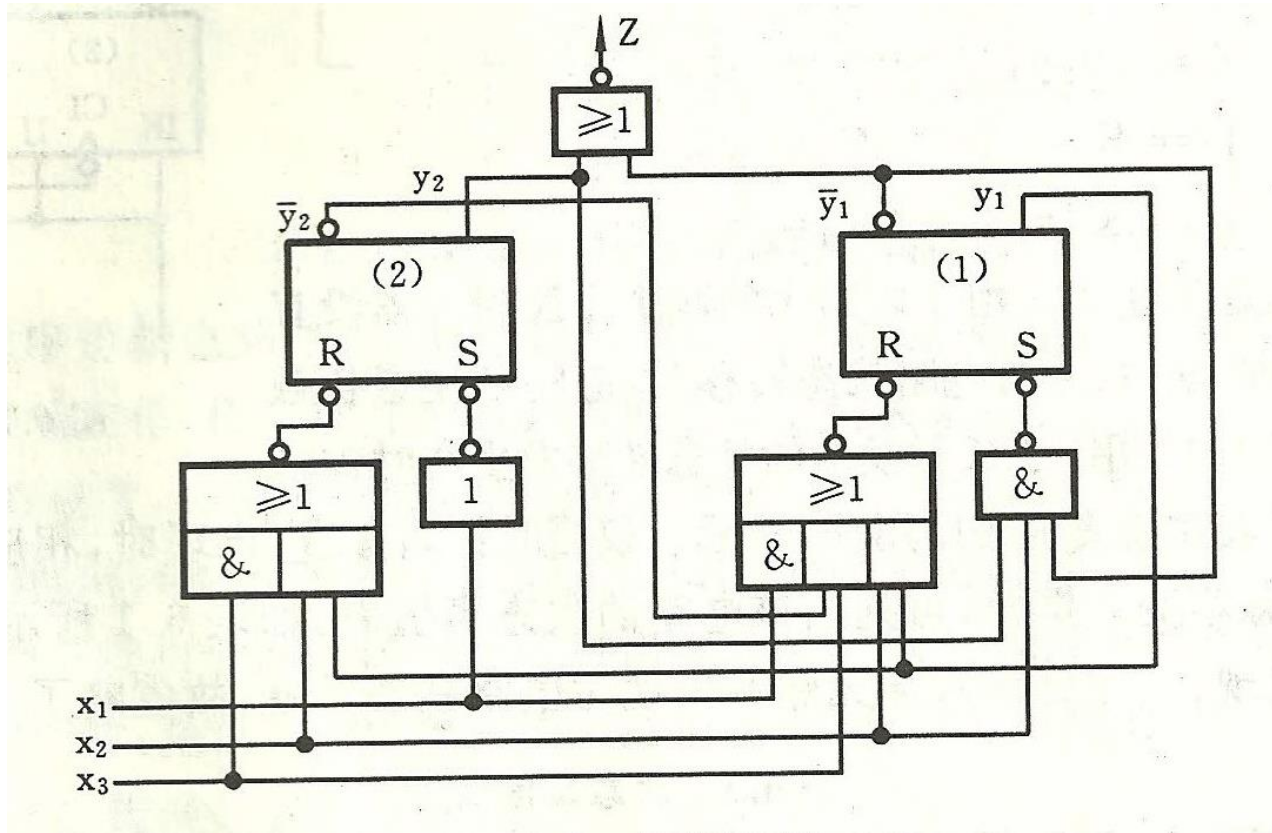
(4) Draw the timing diagram and describe its logic function

In order to further describe the state of the circuit under the action of the input pulse and the output change process, the timing diagram of the circuit can be drawn according to the state table or state as shown in the following figure.。

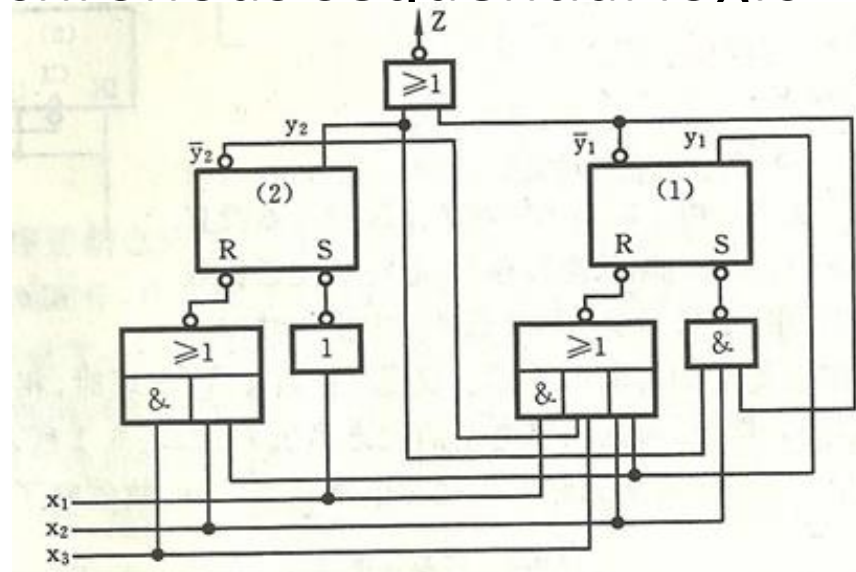


It can be seen from the state diagram and the time diagram that the circuit is a mode 4 incremental counter, and when the fourth input pulse is received, the circuit generates a carry output pulse.

Analysis of pulse asynchronous sequential circuit - Example



- A: The circuit's storage component consists of two basic RS flip-flop. Three inputs, x_1 , x_2 , and x_3 ; one output, Z is the state function of the circuit. Thus, it is a Moor type asynchronous sequential logic circuit.



- 1: List the output function and excitation function:

$$\begin{aligned}
 Z &= \overline{y_2 + \overline{y_1}} = \overline{y_2} y_1 & R_2 &= \overline{x_3 + x_2 y_1} & S_2 &= \overline{x_1} \\
 R_1 &= \overline{x_1 + x_3 \overline{y_2} + x_2 y_1} & S_1 &= \overline{x_2 y_2 \overline{y_1}}
 \end{aligned}$$

$$Z = \overline{y_2} + \overline{y_1} = \overline{y_2} y_1$$

$$R_2 = \overline{x_3 + x_2 y_1}$$

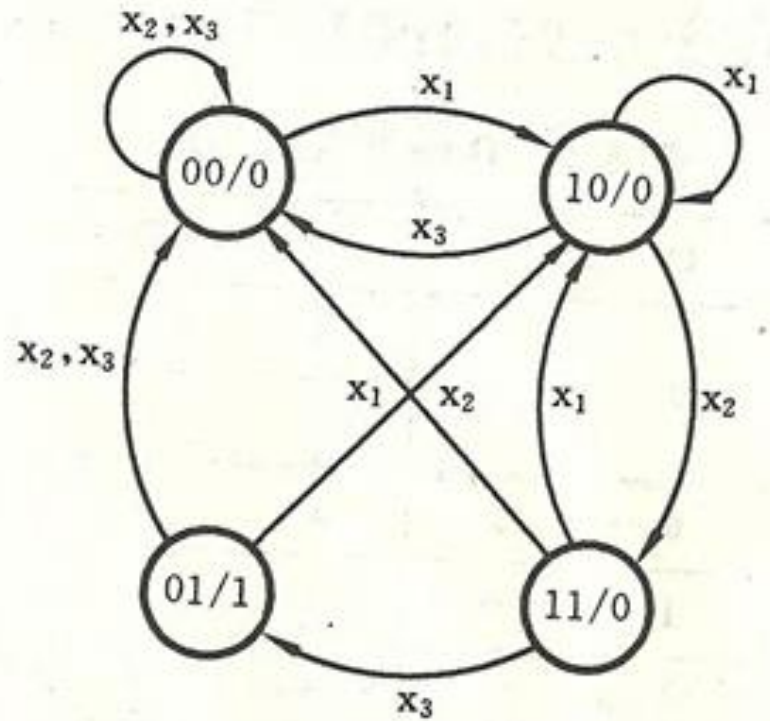
$$S_2 = \overline{x_1}$$

$$R_1 = \overline{x_1 + x_3 \overline{y_2} + x_2 y_1}$$

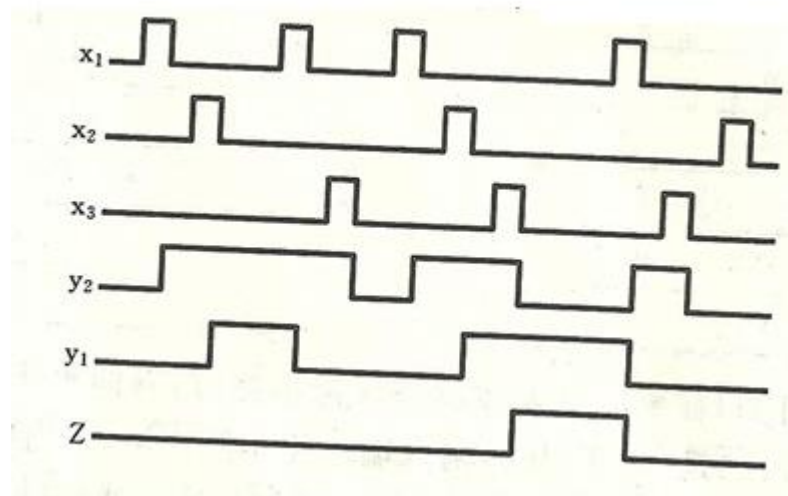
$$S_1 = \overline{x_2 y_2 \overline{y_1}}$$

Input			Present		Excitation				Next state	
x_1	x_2	x_3	y_2	y_1	R_2	S_2	R_1	S_1	y_2^{n+1}	y_1^{n+1}
1	0	0	0	0	1	0	0	1	1	0
1	0	0	0	1	1	0	0	1	1	0
1	0	0	1	0	1	0	0	1	1	0
1	0	0	1	1	1	0	0	1	1	0
0	1	0	0	0	1	1	1	1	0	0
0	1	0	0	1	0	1	0	1	0	0
0	1	0	1	0	1	1	1	0	1	1
0	1	0	1	1	0	1	0	1	0	0
0	0	1	0	0	0	1	0	1	0	0
0	0	1	0	1	0	1	0	1	0	0
0	0	1	1	0	0	1	1	1	0	0
0	0	1	1	1	0	1	1	1	0	1

Present		next	$y_2^{n+1} y_1^{n+1}$			Output Z
			x_1	x_2	x_3	
0	0	10	00	00	00	0
0	1	10	00	00	00	1
1	0	10	11	00	00	0
1	1	10	00	00	01	0



- Draw the timing diagram and explain the logic function. Assume the input sequence is x_1 - x_2 - x_1 - x_3 - x_1 - x_2 - x_3 - x_1 - x_3 - x_2 , and based on the state diagram, the timing diagram is as follows:



6.1.3 Design of Pulse Asynchronous Sequential Logic Circuit

I: Methods and steps

1.Methods: The design method of pulse asynchronous sequential logic circuit is basically the same as the design of synchronous sequential logic circuit. Two problems should be paid attention to.

(1) Since two or more inputs are not allowed to be 1 at the same time (1 indicates that a pulse appears), the design can be handled as follows:

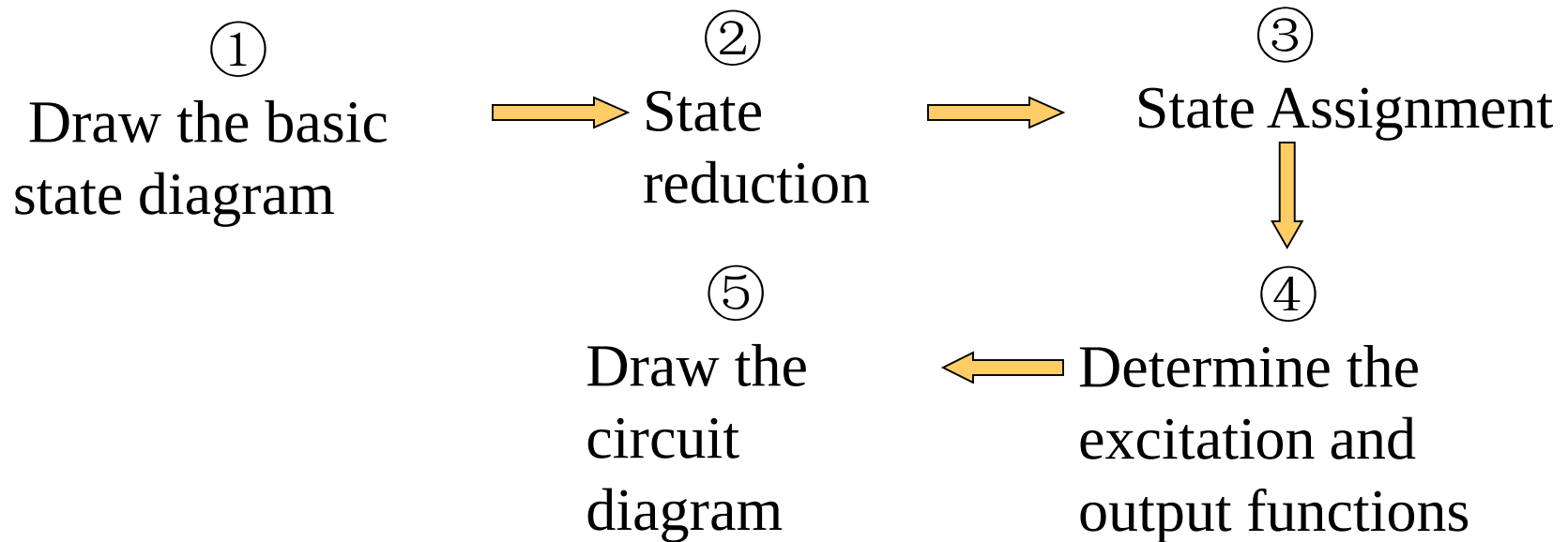
➡ When there are multiple input signals, it is only necessary to consider the case where only one of input signals is one;

➡ When determining the excitation function and the output function, the case where two or more inputs are simultaneously 1 can be handled as a don't care term or condition.

(2) When the storage circuit uses a flip-flop with a CP, the CP of the flip-flop should be treated as an excitation function. The design is facilitated by the integrated processing of the CP and the input of the flip-flop.

2. Steps

The design process is the same as the synchronous sequential circuit, as follows:

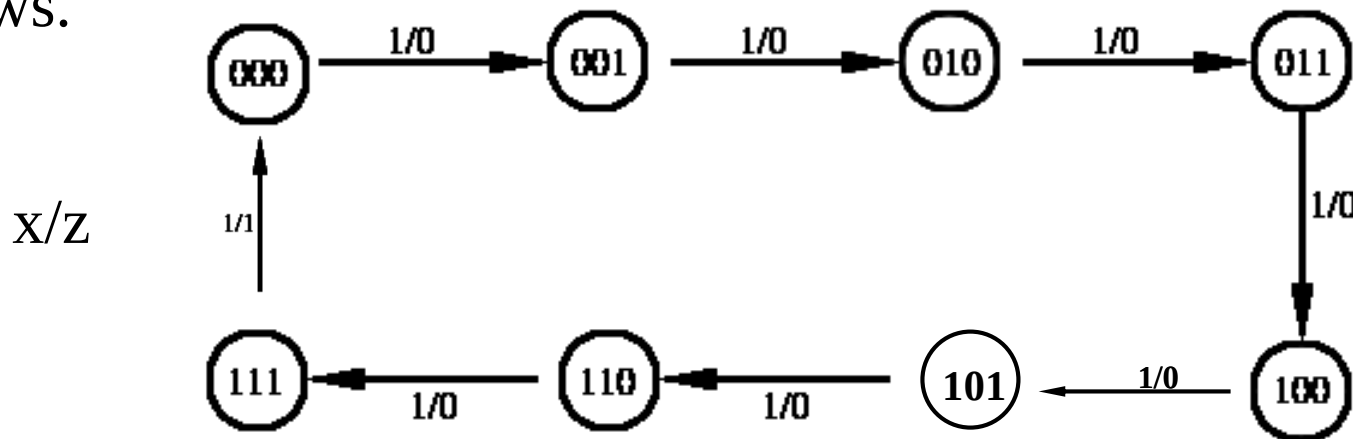


II. Example: Using a T flip-flop as a storage element to design an asynchronous Mode 8 incremental counter. The circuit counts the pulses appearing at the input x. When the eighth pulse is received, the output Z produces a carry output pulse.

Solution: From the meaning of the problem, the circuit model is Mealy type. Since the state binary number and state conversion relationship are very clear, a binary state diagram and a state table can be made directly.

(1) Draw the state diagram and state table

Let the initial state of the circuit be "000", and the state variables be represented by y_2, y_1, y_0 . A binary state diagram can be made as follows.



The corresponding binary status table is:

Present	Next $y_2^{n+1}y_1^{n+1}y_0^{n+1}$ / Output Z
$y_2 y_1 y_0$	$x = 1$
0 0 0	0 0 1 / 0
0 0 1	0 1 0 / 0
0 1 0	0 1 1 / 0
0 1 1	1 0 0 / 0
1 0 0	1 0 1 / 0
1 0 1	1 1 0 / 0
1 1 0	1 1 1 / 0
1 1 1	0 0 0 / 1

(2) Determine the excitation function and the output function

Assuming that the state is unchanged, the CP of the corresponding flip-flop is 0, and the input T is arbitrary. When the state needs to be changed, the CP of the corresponding flip-flop is 1 (with a pulse appearing), and T is 1.

According to the state table, the truth table of the excitation function and the output function when x is 1 can be obtained:

Input x	Present $y_2 y_1 y_0$			Next $y_2^{(n+1)} y_1^{(n+1)} y_0^{(n+1)}$			Excitation $C_2 T_2 C_1 T_1 C_0 T_0$						Output Z
	y_2	y_1	y_0	$y_2^{(n+1)}$	$y_1^{(n+1)}$	$y_0^{(n+1)}$	C_2	T_2	C_1	T_1	C_0	T_0	
1	0	0	0	0	0	1	0	d	0	d	1	1	0
1	0	0	1	0	1	0	0	d	1	1	1	1	0
1	0	1	0	0	1	1	0	d	0	d	1	1	0
1	0	1	1	1	0	0	1	1	1	1	1	1	0
1	1	0	0	1	0	1	0	d	0	d	1	1	0
1	1	0	1	1	1	0	0	d	1	1	1	1	0
1	1	1	0	1	1	1	0	d	0	d	1	1	0
1	1	1	1	0	0	0	1	1	1	1	1	1	1

6.5 D

Q	Q^{n+1}	CP	D
0	0	d	0
		0	d
0	1	1	1
1	0	1	0
1	1	d	1
		0	d

6.7 T

Q	Q^{n+1}	CP	T
0	0	d	0
		0	d
0	1	1	1
1	0	1	1
1	1	d	0
		0	d

6.6 J-K

Q	Q^{n+1}	CP	J	K
0	0	d	0	d
		0	d	d
0	1	1	1	d
1	0	1	d	1
1	1	d	d	0
		0	d	d

6.8 R-S

Q	Q^{n+1}	CP	R	S
0	0	d	d	0
		0	d	d
0	1	1	0	1
1	0	1	1	0
1	1	d	0	d
		0	d	d

According to the excitation function and the output function truth table, and considering that x is 0 (no pulse input, the circuit state is unchanged), the CP of each flip-flop can be 0, and the input T is arbitrary. The simplified excitation function and output function expression can be obtained as follows:

$$C_2 = xy_1y_0 \quad ; \quad T_2 = 1$$

$$C_1 = xy_0 \quad ; \quad T_1 = 1$$

$$C_0 = x \quad ; \quad T_0 = 1$$

$$Z = xy_2y_1y_0$$

(3) Draw the logic circuit diagram

According to the excitation function and the output function expression, the logic circuit that realizes the given requirement can be drawn as shown in the following figure.

